iLearn Practice Exam

10BR · Chapter / Topic 10

Cambridge IGCSE Computer Science 0478

Paper 2 Algorithms, Programming and Logic

Chapter 10: Boolean Logic

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 75

Marks obtained: ..... / 75

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 9 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 Consider the logic expression:

Z is 1 if (**A** = 1 AND **C** = NOT 1) AND (**B** = 1 NOR **C** = 1)

(a) Draw a logic circuit for this logic expression.

Each logic gate must have a maximum of **two** inputs.

Do **not** simplify this logic expression.



[4]

(b) Complete the truth table from the given logic expression.

A	B	C	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

2 There are **three** descriptions of logic gates. Each logic gate has two inputs **A** and **B** with one output **X**.
Identify each logic gate.
Complete a truth table for each logic gate.

(a) The only time the output is 1 is when both inputs are 1.

Logic gate .....

Complete the truth table for this description.

A	B	X
0	0	
0	1	
1	0	
1	1	

[2]

(b) The output is 1 when both inputs are different.

Logic gate .....

Complete the truth table for this description.

A	B	X
0	0	
0	1	
1	0	
1	1	

[2]

(c) The only time the output is 1 is when both inputs are 0.

Logic gate .....

Complete the truth table for this description.

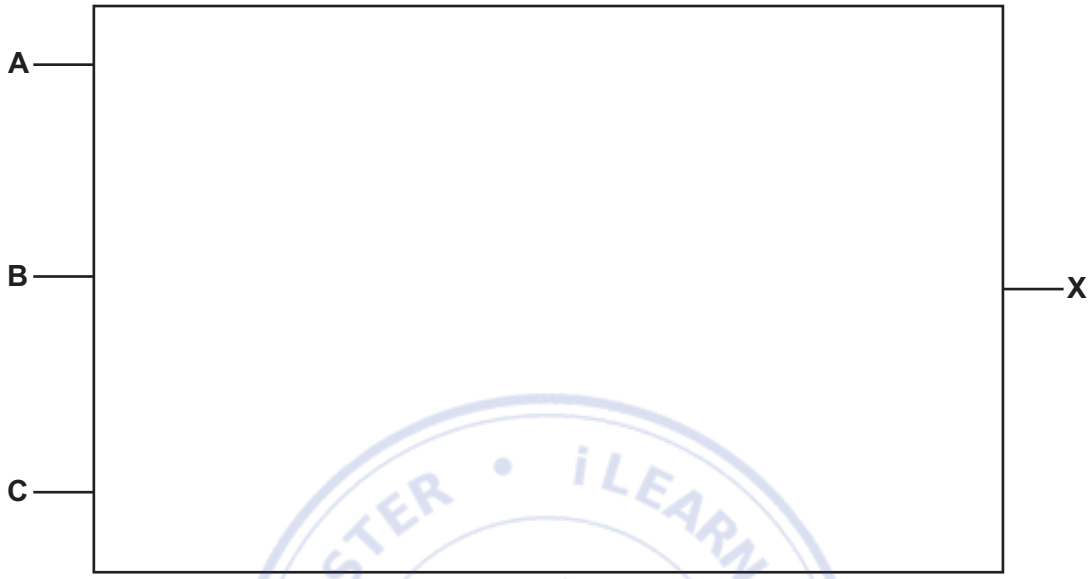
A	B	X
0	0	
0	1	
1	0	
1	1	

[2]

(d) Consider this logic expression:

X = (NOT **A** OR NOT **B**) OR NOT **C**

Draw a logic circuit for this logic expression. Each logic gate must have a maximum of **two** inputs. Do **not** attempt to simplify this logic expression.

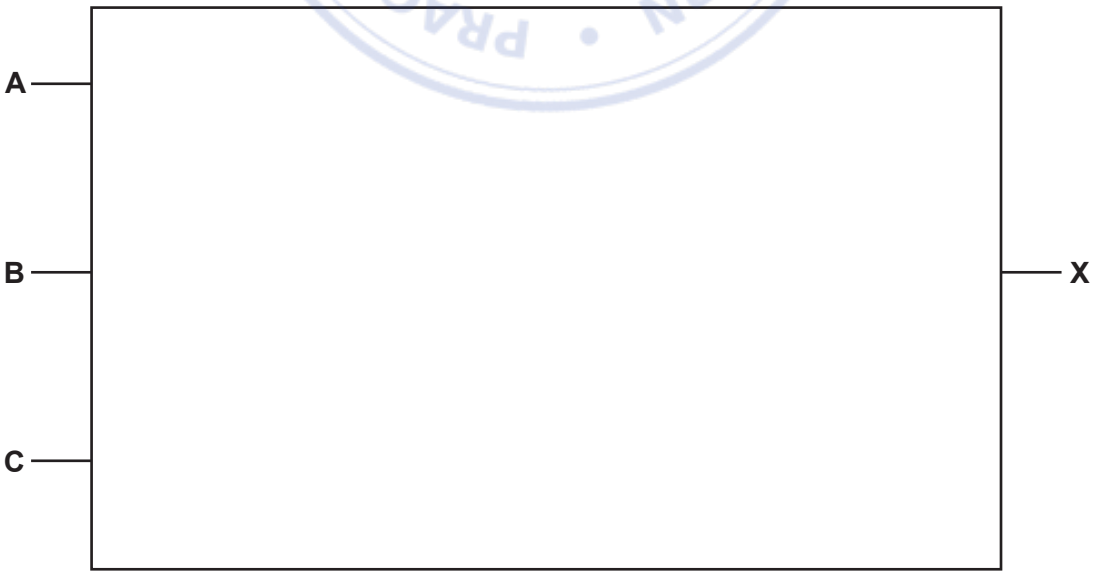


[5]

3 Consider the logic expression:

X = (**A** XOR **B**) AND (NOT **B** AND **C**)

(a) Draw a logic circuit for this logic expression. Each logic gate must have a maximum of **two** inputs. Do **not** simplify this logic expression.



[4]

(b) Complete the truth table from the given logic expression.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

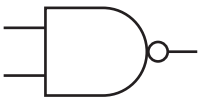
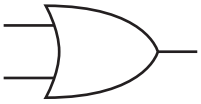
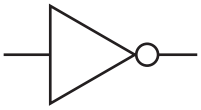
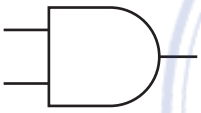
[4]

MASTER · iLEARN CS · LEARN · PRACTISE · iLearn CS

4 **(a)** **Four** logic gate symbols and **five** logic functions are shown.

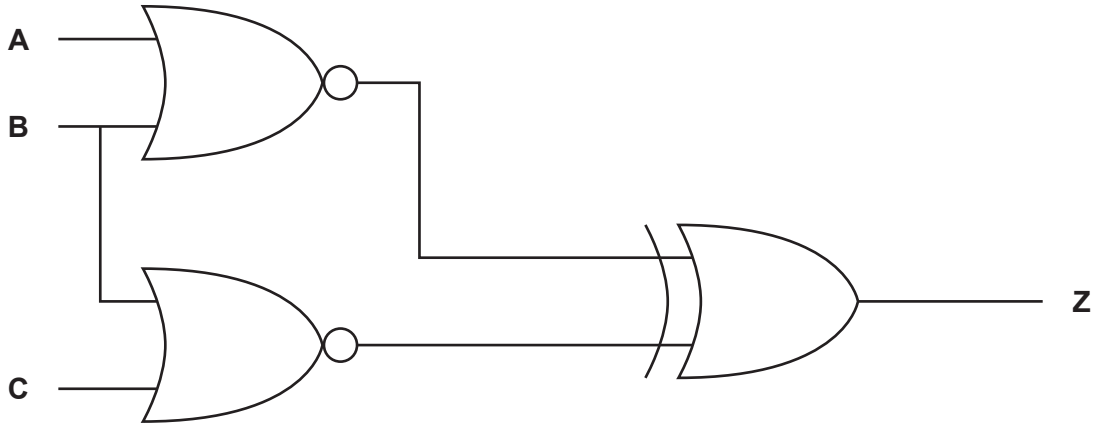
Draw **one** line to link each logic gate symbol to the appropriate logic function.

Not all logic functions will be used.

Logic gate symbol	Logic function
	AND
	XOR
	NOT
	NAND
	OR

[4]

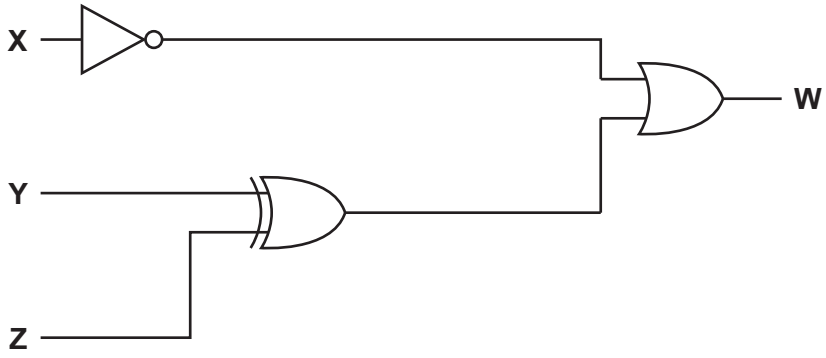
(b) Complete the truth table for this logic circuit.



A	B	C	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

5 Consider the logic circuit:



(a) Write a logic expression for the given logic circuit. Do **not** attempt to simplify the logic expression.

W = ...

..... [3]

(b) Complete the truth table from the given logic circuit.

X	Y	Z	Working space	W
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

6 Consider this logic expression:

Z = (**A** NAND **B**) XOR (NOT (NOT **B** NAND **C**))

(a) Draw a logic circuit for this logic expression.

Each logic gate must have a maximum of **two** inputs.

Do **not** simplify the logic expression.



[5]

(b) Complete the truth table from the given logic expression.

A	B	C	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

7 Consider the logic expression:

X = (**J** XOR NOT **K**) NAND NOT **L**

(a) Draw a logic circuit for the logic expression.
Each logic gate must have a maximum of **two** inputs.
Do **not** simplify the logic expression.



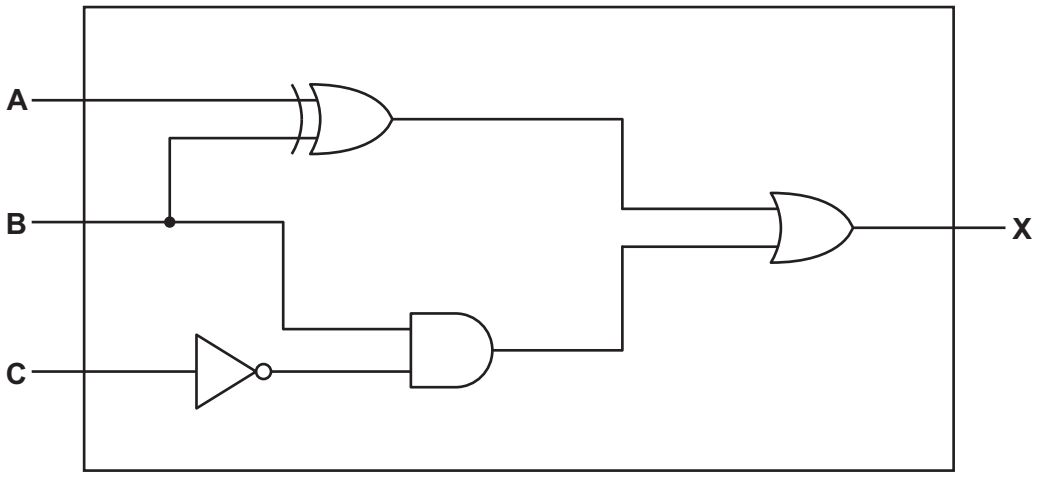
[4]

(b) Complete the truth table for the given logic expression.

J	K	L	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

8 Consider the logic circuit:



(a) Write the logic expression for the given logic circuit.

.....

..... [3]

(b) Complete the truth table for the given logic circuit.

A	B	C	working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

9 Consider the logic expression:

Z = NOT (**B** OR NOT **C**) XOR (**A** NAND **C**)

(a) Draw a logic circuit for this logic expression.

Each logic gate must have a maximum of **two** inputs.

Do **not** simplify the logic expression.



[5]

(b) Complete the truth table from the given logic expression.

A	B	C	working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[4]

END OF QUESTION PAPER

MASTER · iLEARN CS · LEARN · PRACTISE · iLearn CS